



PROGRAMMING 6212

PROG6212

ABSTRACT
POE PART 1

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ERD Race Day System

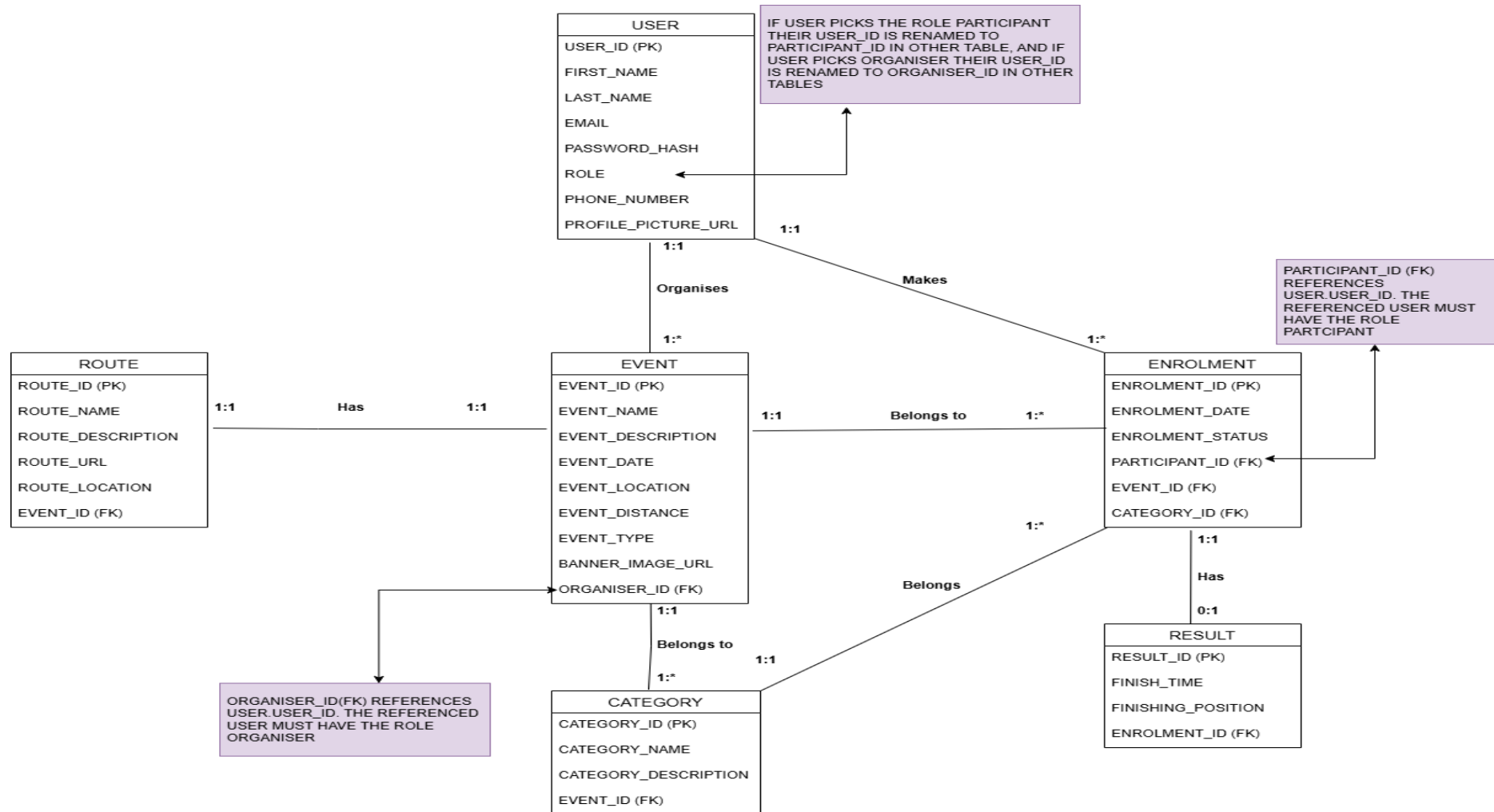


Figure 1 (Coronel, et al., 2020) Entity Relation Diagram

API Endpoint Plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as either an Organiser or Participant.	None	FIRST_NAME, LAST_NAME, EMAIL, PASSWORD, ROLE, PHONE_NUMBER	201 Created – User registered successfully. 400 Bad Request – Invalid details/role. 409 Conflict – Email already exists.
POST	/api/auth/login	Authenticates a user and starts their authenticated session.	None	EMAIL, PASSWORD	200 OK – Login successful and user/session information returned. 401 Unauthorized – Invalid credentials.
POST	/api/auth/logout	Ends the currently authenticated user's session.	Any	None	200 OK – Logged out successfully. 401 Unauthorized – User is not authenticated.
GET	/api/users/profile	Retrieves the profile of the currently loggedin user.	Any	None	200 OK – User profile returned. 401 Unauthorized – Not logged in.
PUT	/api/users/profile	Updates the currently loggedin user's profile information.	Any	FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER	200 OK – Profile updated. 400 Bad Request – Invalid information. 401 Unauthorized – Not logged in.
GET	/api/events	Retrieves a list of available events.	None	None	200 OK – List of events.

GET	/api/events/{id}	Retrieves the details of a specific event.	None	None	200 OK – Event details. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new event for the loggedin Organiser.	Organiser	EVENT_NAME, EVENT_DESCRIPTION, EVENT_DATE, EVENT_LOCATION, EVENT_DISTANCE, EVENT_TYPE	201 Created – Event created. 400 Bad Request – Invalid data. 401 Unauthorized – Not logged in. 403 Forbidden – User is not an Organiser.
PUT	/api/events/{id}	Updates an event belonging to the loggedin Organiser.	Organiser	EVENT_NAME, EVENT_DESCRIPTION, EVENT_DATE, EVENT_LOCATION, EVENT_DISTANCE, EVENT_TYPE	200 OK – Event updated. 404 Not Found – Event not found. 403 Forbidden – User is not authorised.
DELETE	/api/events/{id}	Deletes an event belonging to the loggedin Organiser.	Organiser	None	204 No Content – Event deleted. 404 Not Found – Event not found. 403 Forbidden – User is not authorised.
GET	/api/events/{eventId} /categories	Retrieves all categories available for an event.	None	None	200 OK – List of categories. 404 Not Found – Event not found.
POST	/api/events/{eventId} /categories	Creates a category for an event managed by the Organiser.	Organiser	CATEGORY_NAME, CATEGORY_DESCRIPTOR	201 Created – Category created. 400 Bad Request – Invalid data. 403 Forbidden – User is not authorised.
PUT	/api/categories/{id}	Updates an existing event category.	Organiser	CATEGORY_NAME, CATEGORY_DESCRIPTOR	200 OK – Category updated. 404 Not Found – Category not found. 403 Forbidden – User is not authorised.

DELETE	/api/categories/{id}	Deletes an event category.	Organiser	None	204 No Content – Category deleted. 404 Not Found – Category not found. 403 Forbidden – User is not authorised.
POST	/api/events/{eventId}/enrolments	Allows a Participant to enrol in an event and select a category.	Participant	CATEGORY_ID	201 Created – Enrolment created. 400 Bad Request – Invalid category. 401 Unauthorized – Not logged in. 403 Forbidden – User is not a Participant. 409 Conflict – Already enrolled.
GET	/api/enrolments/my	Retrieves all events in which the logged-in Participant is enrolled.	Participant	None	200 OK – Participant's enrolments. 401 Unauthorized – Not logged in.
GET	/api/events/{eventId}/enrolments	Retrieves all Participants enrolled in an event managed by the Organiser.	Organiser	None	200 OK – List of enrolments. 403 Forbidden – User is not the event's Organiser. 404 Not Found – Event not found.
GET	/api/enrolments/{id}	Retrieves details of a specific enrolment.	Any authenticated user	None	200 OK – Enrolment details. 403 Forbidden – User cannot access this enrolment. 404 Not Found – Enrolment not found.
POST	/api/events/{eventId}/results	Records a Participant's finishing time and position after an event.	Organiser	PARTICIPANT_ID, FINISH_TIME, FINISHING_POSITION	201 Created – Result recorded. 400 Bad Request – Invalid result. 403 Forbidden – User is not authorised.
GET	/api/results/my	Retrieves the logged-in Participant's	Participant	None	200 OK – Participant's results. 401 Unauthorized – Not logged in.

		personal race results.			
GET	/api/events/{eventId}/results	Retrieves results for Participants in an Organiser's event.	Organiser	None	200 OK – Event results. 403 Forbidden – User is not the event's Organiser. 404 Not Found – Event not found.

Figure 2 (Troelsen & Japikse, 2022) API Endpoint Plan

GitHub Commits

Commits · RealebogaMoche25/RaceDay_POE

https://github.com/RealebogaMoche25/RaceDay_POE/commits/main/

RealebogaMoche25 / RaceDay_POE

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Commits

main

All users All time

Commits on Sep 3, 2026

Update README with CI workflow check validation	Verified	5b233a5	
RealebogaMoche25 authored 16 minutes ago · 1 / 1			
Merge branch 'main' of https://github.com/RealebogaMoche25/RaceDay_POE		744d005	
RealebogaMoche25 committed 39 minutes ago · 1 / 1			
improved code structure for readability		d980961	
RealebogaMoche25 committed 41 minutes ago			
Simplify repository check workflow	Verified	8353a84	
RealebogaMoche25 authored 2 hours ago · 1 / 1			
Refactor repository structure validation checks	Verified	3a9fedb	
RealebogaMoche25 authored 2 hours ago · 0 / 1			
Fix repository structure validation script	Verified	4e813c6	
RealebogaMoche25 authored 2 hours ago · 0 / 1			
Fix filename case sensitivity for PROG6212 Part 1.pdf	Verified	0ca2cc7	
RealebogaMoche25 authored 2 hours ago · 0 / 1			

Figure 3 (GitHub, n.d.)

Commits · RealebogaMoche25/RaceDay_POE

https://github.com/RealebogaMoche25/RaceDay_POE/commits/main/

Update ERD file check to use PNG format RealebogaMoche25 authored 2 hours ago · 0 / 1	Verified 26f3256	<>
Upgrade checkout action to version 6 RealebogaMoche25 authored 2 hours ago · 0 / 1	Verified 4b32a1a	<>
Fix repository CI workflow triggers RealebogaMoche25 authored 2 hours ago · 0 / 1	Verified e3a0632	<>
Merge branch 'main' of https://github.com/RealebogaMoche25/RaceDay_POE RealebogaMoche25 committed 3 hours ago · 0 / 1	01ca84b	<>
Add Part 1 planning documents and ERD RealebogaMoche25 committed 3 hours ago	3bfac4c	<>
Add repository structure validation workflow RealebogaMoche25 authored 3 hours ago · 0 / 1	Verified af9e436	<>
Create repository-check.yml RealebogaMoche25 authored 3 hours ago	Verified ec08c18	<>

Commits on Sep 1, 2026

Add sample race results for testing RealebogaMoche25 committed 2 days ago	4996930	<>
Add sample participant event enrolments RealebogaMoche25 committed 2 days ago	45e9fe8	<>
Add categories for each RaceDay event RealebogaMoche25 committed 2 days ago	cf09d1c	<>
Add sample routes and route URL references RealebogaMoche25 committed 2 days ago	1eecaa1	<>

Figure 4 (GitHub, n.d.)

Commits · RealebogaMoche25/RaceDay_POE

https://github.com/RealebogaMoche25/RaceDay_POE/commits/main/

Add sample race results for testing RealebogaMoche25 committed 2 days ago	4996930		
Add sample participant event enrolments RealebogaMoche25 committed 2 days ago	45e9fe8		
Add categories for each RaceDay event RealebogaMoche25 committed 2 days ago	cf09d1c		
Add sample routes and route URL references RealebogaMoche25 committed 2 days ago	1eeca1		
Add sample events with required event information RealebogaMoche25 committed 2 days ago	0cf1a12		
Add sample users for organiser and participant testing RealebogaMoche25 committed 2 days ago	14d0aef		

Commits on Aug 31, 2026

ADDED ROUTE_URL TO MATCH THE ERD RealebogaMoche25 committed 3 days ago	ea41c39		
ALTERED A CATEGORY TABLE ATTRIBUTE TO MATCH ERD. CATEGORY TO CATEGORY_DESCRIPTION RealebogaMoche25 committed 3 days ago	24f056a		
RENAMING SOME ATTRIBUTES IN THE EVENT TABLE TO MATCH ERD SINCE IT IS REQUIRED AND ADDED A BANNER_IMAGE_URL SINCE IT IS PRESENT IN ERD RealebogaMoche25 committed 3 days ago	edcbe12		
CREATED ROUTE TABLE AND IMPLEMENT CONSTRAINTS RealebogaMoche25 committed 3 days ago	1d34f24		

Create RESULT table for participant race results. Add result validation and implement relationship constraints.

Figure 5 (GitHub, n.d.)

Commits · RealebogaMoche25/RaceDay_POE

https://github.com/RealebogaMoche25/RaceDay_POE/commits/main/

Chat

IT IS PRESENT IN ERD	edcbe12	
RealebogaMoche25 committed 3 days ago		
CREATED ROUTE TABLE AND IMPLEMENT CONSTRAINTS	1d34f24	
RealebogaMoche25 committed 3 days ago		
Create RESULT table for participant race results. Add result validation and enrolment relationship constraints	c1c2a62	
RealebogaMoche25 committed 3 days ago		
Create ENROLMENT table for participant event entries. Add enrolment status and registration date tracking	9fe6645	
RealebogaMoche25 committed 3 days ago		
Create CATEGORY table for event participation options. Add category descriptions and event relationship constraints	0893818	
RealebogaMoche25 committed 3 days ago		
Create EVENT table for organiser managed events. Add event types and organiser relationship constraints	f6913f3	
RealebogaMoche25 committed 3 days ago		
Create USER table for system accounts and roles	6d444f1	
RealebogaMoche25 committed 3 days ago		
CREATED PROJECT	1e8ebd8	
RealebogaMoche25 committed 3 days ago		
Initial commit	Verified 6eef283	
RealebogaMoche25 authored 3 days ago		

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Figure 6 (GitHub, n.d.)

CI Workflow

The screenshot displays a GitHub web interface for a repository named 'RaceDay_POE'. The browser address bar shows the URL: https://github.com/RealebogaMoche25/RaceDay_POE/blob/main/.github/workflows/repository-check.yml. The left sidebar shows the file explorer with the following structure:

- Files
 - main (selected)
 - Go to file (search bar)
 - .github/workflows
 - repository-check.yml (selected)
 - Docs
 - README.md
 - RaceDay_POE.slnx

The main content area shows the file 'repository-check.yml' with the following details:

- Repository: RaceDay_POE / .github / workflows / repository-check.yml
- Author: RealebogaMoche25
- Commit: 8353a84 · 1 hour ago
- History (link)
- Code (selected), Blame, 24 lines (18 loc) · 899 Bytes

The workflow file content is as follows:

```
1  name: RaceDay Repository CI
2
3  on: [push, pull_request]
4
5  jobs:
6    repository-check:
7      runs-on: ubuntu-latest
8
9    steps:
10     - name: Checkout repository
11       uses: actions/checkout@v6
12
13     - name: Validate repository structure
14       run: |
15         echo "Checking RaceDay repository structure..."
16
17         test -d "Docs" || { echo "ERROR: Docs folder is missing."; exit 1; }
18         test -f "README.md" || { echo "ERROR: README.md is missing."; exit 1; }
19         test -f "Docs/erdprog.drawio.png" || { echo "ERROR: ERD file is missing."; exit 1; }
20         test -f "Docs/PROG6212 PART 1.pdf" || { echo "ERROR: PROG6212 Part 1.pdf is missing."; exit 1; }
21         test -f "Docs/RACEDAY_POE.sql" || { echo "ERROR: RACEDAY_POE.sql is missing."; exit 1; }
22
23         echo "All required Part 1 files are present."
24         echo "RaceDay repository structure check passed successfully."
```

Figure 7 (GitHub, n.d.)

Workflow runs · RealebogaMoche25/RaceDay_POE

https://github.com/RealebogaMoche25/RaceDay_POE/actions

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Actions New workflow

All workflows

RaceDay Repository CI

Management
Caches
Attestations
Runners
Usage metrics
Performance metrics

All workflows
Showing runs from all workflows

Filter workflow runs

11 workflow runs

	Workflow	Event	Status	Branch	Actor
✓	Merge branch 'main' of https://github.com/RealebogaMoche25...	main	16 minutes ago 16s		...
✓	Simplify repository check workflow	main	Today at 17:10 8s		...
✗	Refactor repository structure validation checks	main	Today at 17:06 7s		...
✗	Fix repository structure validation script	main	Today at 17:02 10s		...
✗	Fix filename case sensitivity for PROG6212 Part 1.pdf	main	Today at 16:55 7s		...
✗	Update ERD file check to use PNG format	main	Today at 16:53 7s		...

Figure 8 (GitHub, n.d.)

YouTube link

<https://youtu.be/PDVLrLdDqpQ?feature=shared>

References

Coronel, C., Morris, S., Crockett, K. & Blewett, C., 2020. *Database Principles: Fundamentals of Design, Implementation, and Management*. 3rd ed. Andover: Cengage Learning.

GitHub, n.d. [Online]

Available at: https://github.com/RealebogaMoche25/RaceDay_POE

Troelsen, A. & Japikse, P., 2022. *Pro C# 10 with .Net 6: Foundational Principles and Practices in Programming*. 11th ed. New York: Apress Media LLC.